

Performance Evaluation of Space Data Centers: DDSSO-Assisted LEO Mega-Constellations

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Abstract

Space data centers (SDCs) have been proposed as an alternative to terrestrial data centers, which face critical challenges in power consumption, water usage, resource waste, and carbon emissions. In particular, the dawn-dusk sun-synchronous orbit (DDSSO) assisted low earth orbit (LEO) mega-constellation, so called DDSSO-LEO, has gained increasing attention. However, the performance of DDSSO-LEO has not been closely evaluated. This study simulates a DDSSO-LEO network and evaluates its performance in terms of latency and task success rate (TSR). We further analyze how the performance gain varies with DDSSO computing capability, orbital altitude, and task size, identifying when DDSSO-LEO SDC is beneficial compared to LEO-only SDC.

I. Introduction

The rapid growth of AI applications and data demand has increased the importance of data centers, while terrestrial data centers face critical issues such as excessive power consumption, water scarcity, and resource depletion. As an alternative, space data centers have been proposed by deploying computing resources on satellites [1]. Google and SpaceX both suggested dawn-dusk sun-synchronous orbit (DDSSO) as a promising orbit for this purpose [2], [3]. Since DDSSO enables stable solar power generation, a space data center can be realized as a DDSSO-assisted low earth orbit (LEO) mega-constellation, DDSSO-LEO, where DDSSOs provide high-performance computing and LEO mega-constellations support global connectivity with low-performance computing.

However, the performance of space data centers using DDSSOs has not been sufficiently investigated. In particular, the impacts of DDSSO computing capability, altitude, and task size on user QoS remain unclear. In this paper, we analyze the performance of DDSSO-LEO and provide a preliminary analysis for designing future space data center architectures.

II. System model

In this study, the LEO mega-constellation is modeled based on the Gen2 Starlink constellation, while the DDSSO altitude varies from 400 km to 1200 km in 200 km increments. Tasks are randomly generated at 144 Starlink gateways, and two task-size scenarios are considered: small-size tasks of 10–20 KB and large-size tasks of 1–2 MB. The feasible satellite set for each gateway is determined using the elevation angle and updated every 15 s following the Starlink specification. In addition, the performance is evaluated under different conditions of DDSSO.

A. Feasible set

We set a whole set of LEO satellites (LEOS) and DDSSO satellites (DDSSOS) as follows, $V_L = \{v_1, v_2, \dots, v_L\}$, $V_D = \{v_1, v_2, \dots, v_D\}$. Each gateway k can access only a subset of LEOS and DDSSOS whose elevation angle lies in $25^\circ \leq \theta_{k,v}^\psi \leq 90^\circ$.

Thus, we define each feasible set of LEOS and DDSSOS as follows, respectively,

$$V_{k,L,c}^\psi = \{v_l | v_l \in V_L, 25^\circ \leq \theta_{k,v_l}^\psi \leq 90^\circ\} \quad (1)$$

and

$$V_{k,D,c}^\psi = \{v_d | v_d \in V_D, 25^\circ \leq \theta_{k,v_d}^\psi \leq 90^\circ\} \quad (2)$$

B. Computing offloading model

For a task generated by gateway k at slot ψ , we define the binary offloading variables as $a_{k,v}^\psi, b_{k,v}^\psi$. Each variable is 1 if the gateway k offloads to LEOS and DDSSOS respectively, and otherwise 0.

In this paper, we consider a naive rule-based offloading policy

where gateway k offloads DDSSOS whenever DDSSOS is feasible. Also, we define N_{task} as the total number of generated tasks and $\mathcal{S}$ as the set of successfully served tasks.

C. Computing frequency

The computing frequency is defined as below,

$$f_{D,c,max} = \eta \cdot f_{L,c,max}, \quad (3)$$

where $f_{L,c,max}$ and $f_{D,c,max}$ represent the maximum computing frequency of LEOS and DDSSOS, respectively, and $\eta \geq 1$ denotes the computing frequency ratio.

D. Evaluation Metrics

To evaluate the QoS of the proposed network, we define the latency of the task generated by gateway k at slot ψ as

$$t_k^\psi = t_{\tau,k}^\psi + t_{p,k}^\psi + t_{c,k}^\psi, \quad (4)$$

where $t_{\tau,k}^\psi, t_{p,k}^\psi$, and $t_{c,k}^\psi$ denote the transmission delay, propagation delay, and computing latency, respectively.

We evaluate the performance using three metrics: average latency, L99 latency, and task success rate (TSR). The average latency is defined as

$$T_{\text{avg}} = \frac{1}{|\mathcal{S}|} \sum_{(k,\psi) \in \mathcal{S}} t_k^\psi. \quad (5)$$

The L99 latency is defined as the 99th percentile of the latency values, i.e.,

$$T_{\text{L99}} = \text{Percentile}_{99} \left(\{t_k^\psi\}_{(k,\psi) \in \mathcal{S}} \right). \quad (6)$$

The TSR is defined as the ratio of the number of successfully served tasks to the total number of generated tasks:

$$P_{\text{fea}} = \frac{|\mathcal{S}|}{N_{\text{task}}}. \quad (7)$$

III. Simulation Results

In this study, we evaluate the underlying performance of a DDSSO-LEO using a naive computation offloading policy. Specifically, each task is offloaded to a DDSSO whenever at least one DDSSO is feasible; otherwise, the task is offloaded to a feasible LEOS. We evaluate the network performance with respect to three metrics.

Fig. 1 (a) presents the average latency according to the DDSSO altitude and the computing frequency ratio η . The results show that the latency gain obtained by using DDSSOs strongly depends on η . When $\eta = 5$, the DDSSO-LEO can either outperform or underperform the LEO baseline depending on the DDSSO altitude. In contrast, when $\eta \geq 10$, DDSSO-LEO consistently outperforms the LEO baseline, regardless of the altitude. When $\eta \leq 3$, DDSSO-LEO achieves latency comparable to the LEO baseline only at low

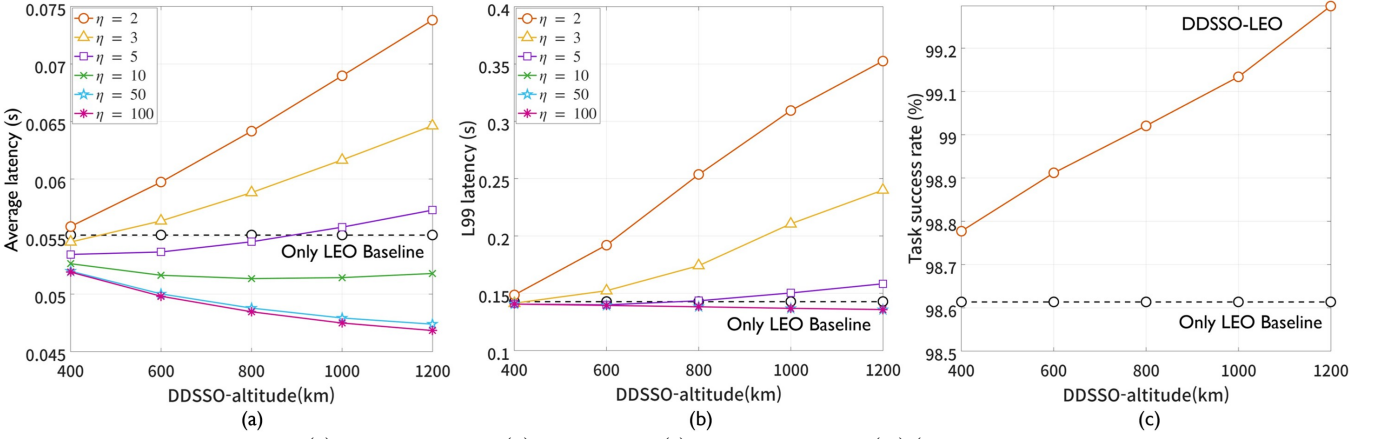


Figure 1. Evaluation results (a) Average latency (b) L99 latency (c) Task success rate (%) (the black dashed line indicates the baseline performance of the task offloading scheme solely with the LEO satellites).

altitudes. This is because the additional computing capability of DDSOs is not sufficient to compensate for the increased propagation delay at higher altitudes.

However, the marginal latency gain becomes limited beyond a certain value of η . In particular, the latency reduction from $\eta = 5$ to $\eta = 10$ is comparable to that from $\eta = 10$ to $\eta = 50$. This implies that increasing the DDSO computing frequency does not always provide proportional latency improvement, since other delays become dominant once the computing latency is sufficiently reduced.

Fig. 1 (b) shows the L99 latency performance. Similar to the average latency results, the L99 latency decreases as the computing frequency ratio increases. This indicates that higher DDSO computing capability can also improve the tail latency performance. However, when η becomes larger than 5, the additional L99 latency gain becomes gradually limited. Therefore, excessively increasing the computing frequency may not be efficient.

Fig. 1(c) shows the TSR, which represents the fraction of tasks that can be served through satellite connectivity. In the considered LEO mega-constellation with an inclination of 53° , some tasks located in high-latitude regions cannot be served by the LEO only. By jointly using DDSOs in polar orbits, the TSR increases because they provide additional coverage opportunities. In particular, as the DDSO altitude increases, the coverage area becomes wider, allowing more high-latitude tasks to be served. This result indicates that DDSOs can improve the service TSR of the LEO mega-constellation, especially for tasks located at high latitudes.

We present the representative results for the small-task case, as similar trends were observed across different task sizes. However, the performance gain of DDSOs becomes larger as the task size increases since larger tasks are more sensitive to computing latency and can better exploit the high computing capability of the DDSO layer. As a result, DDSO-assisted LEO mega-constellations become a plausible architecture for large-size tasks requiring high computing capability.

IV. Conclusion and Discussion

The simulation results demonstrate that DDSO computing frequency, DDSO altitude, and task size are all key factors that affect the network performance of the DDSO-LEO. As the DDSO altitude increases, the coverage area becomes wider and more tasks can be served. However, a higher altitude also increases the propagation delay. Therefore, DDSOs can outperform the LEO baseline only when their computing frequency is sufficiently high to compensate for the additional propagation delay.

The results also show that the benefit of increasing the DDSO computing frequency is limited beyond a certain point. Once the computing latency becomes sufficiently small, the total latency is mainly determined by the propagation delay. This indicates that

simply increasing the computing resources of DDSOs is not always efficient, and thus, an appropriate computing frequency and orbital altitude should be jointly selected according to the target service.

In addition, the task size has a significant impact on the effectiveness of DDSO-assisted offloading. For small tasks, the propagation delay can dominate the total latency, and therefore offloading to DDSOs may not always be beneficial. For larger tasks, however, the computing latency becomes more important, and the DDSO layer can provide a larger performance gain. This implies that the offloading decision between LEOS and DDSO should be determined according to the service requirement, rather than using a fixed-rule based policy.

This study provides an underlying performance evaluation of a DDSO-LEO under a naive offloading policy. To further improve network performance, software-defined networks (SDN) should be introduced into the multi-layer DDSO-LEO architecture. Through SDN-based control, task offloading, path selection, and resource allocation can be jointly optimized according to task size or service type. This can prevent bottlenecks and improve the overall performance of the multi-layer satellite network.

For this purpose, an SDN controller capable of managing the mobility of different orbital layers is required [5]. Since both the LEO and DDSO layers are highly dynamic, the controller should support topology-aware path selection and resource allocation across heterogeneous orbital layers. In addition, practical deployment issues such as the launch cost of DDSOs, altitude-dependent radiation effects, and thermal management should be jointly investigated in future work.

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